

# Work In Progress: Integrating the RF Characteristics of RFID into Undergraduate EE Courses

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**Abstract** - Radio Frequency Identification is becoming important for the tracking of goods and capital items, as well as being used for sensor networks to send data. A reader, which is a receiver and transmitter, sends an interrogating signal to the tag. The tag, be it active or passive, responds and the response, detected by the reader is decoded and transferred to the users IT systems. There are several frequency ranges for tags, ranging from the low kHz, to the ISM frequency at 13.56 MHz to tags operating in the Part 15 and ISM bands near 900 and 2400 MHz. In addition, RFID tags may be active or passive and may be permanently programmed or field re-writable. Each of the classes of tags has fundamental engineering limitations, based upon size, frequency range, and/or data requirements. There are also many design choices for readers based on whether the reader is to be fixed or handheld. The purpose of this work-in-progress is to outline how the RF design consideration effect RFID choices and how the design of RFID systems is being included in courses on Electromagnetics and Communications, both in terms of system design and in design of individual components. Design of antennas for the tag and reader, as well as, an understanding of the fundamental limits based upon frequency choice can be included in an undergraduate electromagnetics course. System design, including link budget, modulation type and overall system parameters can be included as good example in a Communications course. This work-in-progress will also outline chip requirements for the tags and a discussion on how specific cost requirements for tags influence the designs.

**book.**

*Index Terms* – Electromagnetics, RFID, Sensor Networks, Undergraduate Electrical Engineering Curriculum

## INTRODUCTION

Radio Frequency Identification (RFID) Systems are becoming ubiquitous. Almost everywhere one shops, there are RFID tags that are either used for theft protection or for identification of products. In the backroom or warehouse RFID systems are being used to track products. Further RFID is also being used to track assets. Almost daily one reads about new and novel applications of RFID systems including animal identification, runner tracking in marathons, and using a credit card without swiping it through a reader. Components throughout the

supply chain are being labeled with RFID tag to track both position and history.

Many of the applications of RFID involve linking the data to an enterprise system. While this important in the general discussion of RFID, the purpose of this work is to describe how concepts of RFID can be in undergraduate Electrical Engineering courses. RFID provides a rich set of examples to illustrate various topics as well as to show the relevance of these topics to current hot product.

In the next section RFID systems will be described. In subsequent section RFID systems are linked to topics in the Electrical Engineering Curriculum.

## RFID SYSTEMS

Engineering wise, RFID systems may be described as a communications system with the tag classified as a repeater and the reader, a transmitter/receiver pair. The reader transmits a signal, which is received by the tag and reflected or retransmitted with additional information.

An RFID tag may be classified as active or passive, depending whether or not it contains a transmitter, or just acts in a reflective mode. Tags may be classified by the number of information bits transmitted to the reader, with single bit tags used most often for theft deterrence and multibit tags used for data applications. There are several international frequency bands where RFID may be used. An overview of general RFID systems can be found in [1] and UHF system, more specifically in [2] A several part Radio-Oriented Introduction appears in [3].

One overarching requirement of any RFID system is cost. For tags used to monitor high value items such as cargo containers, the cost of the tag is minimal, compared to its value, thus the cost of the individual tag never enters into the analysis. On the other hand, there are many applications where to remain competitive the additional of an RFID Tag to the product must be measured in cents for the used of RFID to be effective. Thus, regardless of other engineering considerations, RFID systems may be used as examples of cost-sensitive designs.

## RFID AND EE TOPICS

In this section is a non-inclusive list of topics that RFID could be used in the EE curriculum.

### 1. Link Budgets

Since RFID is a communications system, the link budget can be written for the system. While many manufacturers only vaguely describe the system specifications in their brochures, it is instructive to show how, by writing a link budget using specifications and standards, the performance of the system can be predicted and compared to the manufacturer's claims.

### II. Near vs. Far and Propagation

Since RFID systems operate at frequencies from 130 kHz to 2.4 GHz, having a reader-to-tag separation of several meters results in being in the near field and some frequencies and the far field at other frequencies. This broad range of possible frequencies allows the introduction of how short range propagation can affect a system.

Because RFID is used in a variety of environments from in the outdoors to inside a store or warehouse, the effects of propagation can be important and useful to discuss. In addition, several frequencies are better to use near metal items or when the tags are attached to metal objects.

### III. Antenna Design

Since except for possibly RFID systems operating at 2.4 GHz, the tag (and most likely the reader) will be much smaller than a quarter wavelength. Thus, antenna and matching system design are a very important part of tag and reader design. Indeed many tag designs use magnetically coupled loops whose operation more closely mimics a transformer than an antenna system

### IV. Novel Fabrication

Often the tag itself is constructed using metalized paper or foil as a substrate. The antenna is die cut into the paper and metallic traces are placed to where the chip will be placed. The chip is placed and glued to the traces using a conductive glue resulting in a cost effective product.

### V. Evaluating Different Technologies

Since RFID systems are really a broad range of products, they encompass various technologies. Some use a reflective (or backscatter) method of returning the answer to in interrogation. Others use a frequency doubler that basically returns a signal at twice the frequency. Finally, others use an active integrated circuit including a transmitter to return the signal.

A second division is active versus passive. While passive RFID tags need no power, active tags require a battery. More novel among these tags are ones that use the power from the interrogation pulse itself to power device storing it in a capacitor.

### VI. Use of International Standards and Standard-based Design

Although RFID tags and systems are relatively simple devices, to ensure compatibility among vendors a large number of standards for RFID tags and systems have been developed. Ideally any vendor's product adhering to the standards should

be able to be read by any other vendor's tag of the same standard.

### VII. Fabless Fabrication

A recent innovation in RFID is to use a 4 color printing press technology to fabricate the chips in the RFID tags [4]. Thus, instead of using a clean room and its associate fabrication processes, simple transistors and their accompanying circuits may be printed. Although the technology is in its infancy, when the process matures, it will open up a new era in low cost circuitry that will be applicable to many areas.

### VIII. Sensor Networks

An exciting extension of RFID is to use have some property be read back rather than just identification data. For example a refrigerated container might read back the current temperature and whether or not the temperature rose above a certain level since the last reading. One can easily envision a broad spectrum of uses for this type of tag and an extension to sensors and networked sensors. For example if one could develop an implantable blood sugar monitor, then using a tag reader, a diabetic could easily monitor his/her blood sugar levels without the current inconveniences

## SUMMARY

The above list is neither all inclusive of the topics that an RFID can be used to illustrate or discuss. RFID systems can be used to illustrate and motivate many topics in the current curriculum, especially in areas of electromagnetics where frequently students do not see the relevance to the material to current technology.

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## REFERENCES

- [1] Finkenzeller, K., *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards and Identification*, second edition, J.F. Wiley, Chichester, England, 2003.
- [2] Glidden, R, et al, "Design of Ultra-Low-Cost UHF RFID Tags for Supply Chain Applications", *IEEE Communications Magazine*, August, 2004, pp140-151.
- [3] Dobkin, D., and T. Wandinger, "A Radio-Oriented Introduction to RFID - Protocols, Tags, and Applications, *High Frequency Electronics*, June and August 2005.
- [4] Jackson, D., "A Fabulous Way to Go Fabless", *Mobile Radio Technology*, March 2006, pp 22-25.